Scientific Report Guidelines

The scientific report serves as a comprehensive document detailing the process, methodology, analysis, and outcomes of the project. It should present findings logically, supporting arguments with relevant data, citations, and clear explanations.

Report Structure and Content Requirements:

1. **Title Page**
    - Include the project title, student names, course name, institution, date, and any other relevant information.
2. **Abstract**
    - Provide a concise summary (150–250 words) of the research question, methodology, key findings, and significance of the results.
3. **Introduction**
    - Introduce the research problem, including context and motivation. Define the specific objectives and hypotheses. Briefly state the methodology and expected contributions.
4. **Literature Review**
    - Summarize key studies and methodologies related to your project topic. Clearly relate past findings to your research question. Identify gaps that your project aims to fill.
5. **Methodology**
    - Describe the simulation or computational methods used (e.g., FEM, FDM, DEM, etc.). Detail the tools, software, and parameters involved. Include any custom algorithms, models, or assumptions. Provide enough technical detail to enable replication.
6. **Results**
    - Present results clearly, using tables, graphs, and figures where appropriate. Use descriptive captions for all visuals and reference them in the text. Include statistical analyses or error assessments as necessary.
7. **Discussion**
    - Interpret the results, explaining their significance relative to the research question. Compare findings with those in the literature. Discuss any unexpected outcomes, limitations, and potential biases.
8. **Conclusion**
    - Summarize main findings and their implications. Suggest future research directions or practical applications. Reflect on the project's overall contribution to the field.
9. **References**
    - Follow a consistent citation style, as outlined in your template. Ensure all sources are cited properly using \cite{}. Only include sources referenced in the text.
10. **Appendices**
    - Add supplementary material, detailed calculations, or additional figures here.

11. Report Structure and Content Requirements:

LaTeX-Specific Do’s and Don’ts

Do’s

- Use Appropriate LaTeX Packages: Include relevant packages for figures (graphicx), tables (tabularx), and citations (biblatex or natbib).
- Structure with Section Headings: Use \section, \subsection, and \subsubsection to create a clear hierarchy and improve readability.
- Cross-Referencing: Use \label{} and \ref{} to link to sections, figures, tables, and equations. This keeps references consistent and automatically updates numbering.
- Figures and Tables: Place figures and tables with \begin{figure} ... \end{figure} and \begin{table} ... \end{table}, positioning them appropriately with [h], [t], etc. Add captions and cite in the text using \ref.
- Bibliography Management: Use BibTeX or BibLaTeX with .bib files for consistent citation formatting. Organize references alphabetically or by appearance, per template guidelines.
- Equation Formatting: Use \begin{equation} ... \end{equation} for standalone equations. Refer to equations in text using \eqref{}.

Don’ts

- Avoid Hard-Coding Page Breaks: Don’t use \newpage or \pagebreak to control page breaks manually. Let LaTeX handle layout adjustments.
- Don’t Overuse Bold/Italic Fonts: Use emphasis sparingly. Stick to \textbf{} for headings if required by the template, but avoid excessive styling within the text.
- Avoid Low-Resolution Images: Use vector formats (e.g., .eps, .pdf) for graphs and high-resolution .png or .jpg for images to maintain quality.
- Avoid Excessive Line Breaks in Text: Don’t use \ to control line breaks within paragraphs; let LaTeX handle text flow naturally.
- Don’t Ignore Template Requirements: Ensure adherence to the given LaTeX template for consistency in font size, margin, citation style, and other formatting specifications.
- Avoid Cluttered Graphs and Tables: Avoid overly dense or complex visuals. Keep graphics clear, with readable labels and legends.

General Do’s and Don’ts in Scientific Writing

Do’s

- Maintain Formal Tone: Write in the third person and use formal, precise language.
- Proofread Carefully: Check for grammar, spelling, and technical accuracy. A well-written report is easier to understand and more professional.
- Keep it Concise: Be clear and to the point. Avoid unnecessary jargon and lengthy explanations of well-known methods.

- Focus on Consistency: Maintain consistent terminology, units of measurement, and formatting throughout.
- Check Citations: Verify that all references are accurate and cited in the correct format.
- Review for Clarity: Ensure that figures, tables, and equations are well-explained and serve a purpose in the text.

Don’ts

- Don’t Plagiarize: Always paraphrase or directly quote with proper attribution when using others' work.
- Avoid Vague Statements: Scientific writing requires precision. Avoid subjective language like “we believe” or “this might mean.”.
- Don’t Overuse Abbreviations: Define abbreviations when first introduced, and use sparingly to avoid confusion.
- Don’t Omit Critical Details: Ensure that the methodology is clear and thorough enough for replication.
- Don’t Skip Data Analysis: Present a reasoned analysis of results rather than leaving interpretation to the reader.